

Creation of a Robust and Generalizable Machine Learning Classifier for Patient Ventilator Asynchrony

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Online Supplement

Reproduction:

Reproduction of our experiment can be performed using our code found at:

https://github.com/hahnicity/ucdpv_pva_ml

Reproduction will require our dataset as well. Please contact JYA for access to the dataset at jyadams@ucdavis.edu.

Supplemental Tables/Figures

Table S1: List of all metadata variables along with a description. These variables were all processed from raw ventilator waveform data and were evaluated as independent features to add to our PVA detection model.

Variable Name	Units	Description
TVi	Milliliters/second	inspiratory tidal volume, defined as the integral of the flow-time curve values from breath start (BS) to point where flow crosses 0 (x0).
TVe	Milliliters/second	expiratory tidal volume, defined as the integral of the flow-time curve values from x0 to breath end (BE)
TVe/TVi	Unitless	the ratio of expiratory tidal volume to inspiratory tidal volume
I-time	Seconds	the time from BS to [x0 minus 1 time point]
E-time	Seconds	the time from x0 to BE
I:E ratio	Unitless	the ratio of the I-time to the E-time
RR	Unitless	instantaneous respiratory rate, defined as 60/breath time
PIF	Liters/minute	peak inspiratory flow, defined as the maximum positive flow recorded from BS to [x0 minus 1 time point]
PEF	Liters/minute	peak expiratory flow, defined as the most negative flow recorded from x0 to BE

PIP	cm H2O	peak inspiratory pressure, defined as the maximum recorded pressure from BS to [x0 minus 1 time point]
Mean flow from PEF	Milliliters	The mean flow observation from the point in time PEF occurred to the point where the breath terminated and a new one begins
ipAUC	cm H2O	the inspiratory pressure area under the curve, defined as the integral of the pressure-time curve from BS to [x0 minus 1 time point]
epAUC	cm H2O	the expiratory pressure area under the curve, defined as the integral of the pressure-time curve from x0 to BE
PEEP	cm H2O	positive end-expiratory pressure, defined as the average of the last 5 data points from the pressure-time curve of each breath
Paw	cm H2O	mean airway pressure
Dynamic Compliance (C_{dyn})	Unitless	The pulmonary compliance of the lung at any given point during a breath. This measure is derived via : $C_{dyn} = \frac{TV_i}{PIP-PEEP}$

Table S2: Descriptive statistics for the all classifiers run on the multiclass classification problem using RUS. ERTC: Extremely Randomized Trees classifier. GBC: Gradient Boosting classifier. MLP; Multi-layer Perceptron. RUS; Random Under-Sampling. DTA; Double-Trigger Asynchrony. BSA; Breath Stacking Asynchrony

Algorithm	Class	Accuracy	Sensitivity	Specificity
Ensemble	Non-PVA	0.9473	0.9315	0.9888
	DTA	0.9467	0.9801	0.9439
	BSA	0.9679	0.9098	0.9823
ERTC	Non-PVA	0.7603	0.7323	0.8336
	DTA	0.8743	0.9973	0.8639
	BSA	0.8062	0.5685	0.865
GBC	Non-PVA	0.9519	0.9382	0.9881
	DTA	0.9534	0.9761	0.9515
	BSA	0.9726	0.9274	0.9838
MLP	Non-PVA	0.9315	0.9162	0.9716
	DTA	0.9391	0.9721	0.9363
	BSA	0.9518	0.8693	0.9723

Table S3: Descriptive statistics for the all class ratios run on the multiclass classification using SMOTE and our ensemble classifier. DTA; Double-Trigger Asynchrony. BSA; Breath Stacking Asynchrony

SMOTE Ratio	Class	Accuracy	Sensitivity	Specificity
0.1	Non-PVA	0.9738	0.9763	0.9672
	DTA	0.9798	0.9069	0.9859
	BSA	0.9783	0.9512	0.985
0.2	Non-PVA	0.9731	0.9747	0.969
	DTA	0.9794	0.9229	0.9842
	BSA	0.9781	0.9476	0.9856
0.3	Non-PVA	0.9719	0.9724	0.9705
	DTA	0.9774	0.9362	0.9808
	BSA	0.9785	0.9435	0.9872
0.4	Non-PVA	0.9716	0.9707	0.9739
	DTA	0.9753	0.9415	0.9781
	BSA	0.9784	0.9414	0.9875
0.5	Non-PVA	0.9713	0.9699	0.975
	DTA	0.9753	0.9415	0.9781
	BSA	0.9787	0.9445	0.9872
0.6	Non-PVA	0.9708	0.969	0.9754
	DTA	0.9747	0.9388	0.9777
	BSA	0.9786	0.9455	0.9868
0.7	Non-PVA	0.97	0.9678	0.9757
	DTA	0.9732	0.9521	0.975
	BSA	0.9782	0.9383	0.9881
0.8	Non-PVA	0.9704	0.9676	0.9776
	DTA	0.9736	0.9495	0.9756
	BSA	0.9787	0.9429	0.9875
0.9	Non-PVA	0.9705	0.9676	0.978
	DTA	0.9742	0.9521	0.976
	BSA	0.9786	0.9435	0.9873
1.0	Non-PVA	0.971	0.9673	0.9806
	DTA	0.9742	0.9601	0.9754
	BSA	0.9793	0.9445	0.9879

